

Critical Value Bounds via the Fiber–Bridge Identity: A Pandrosion Laguerre Inequality

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April 2026

Abstract

Classical Laguerre-type inequalities bound the logarithmic derivative $|P'(z)/P(z)|$ at regular points of a polynomial. At critical points ($P'(c) = 0$), these bounds vanish and give no information on the critical value $P(c)$. We prove a new inequality that fills this gap using the *fiber–bridge identity*: the partial-fraction decomposition of $1/P$ at the fiber level of the branched covering $z \mapsto P(z)$. Combined with orthogonality constraints from the *fiber vanishing identity*, the Cauchy–Schwarz inequality yields a non-trivial lower bound on $|P(c)|$ in terms of root separations. The bound is sharp for $d = 3$.

1 Introduction

Let $P(z) = \prod_{j=0}^{d-1} (z - \alpha_j)$ be a monic polynomial of degree d with roots $\alpha_0, \dots, \alpha_{d-1}$, and let c be a critical point: $P'(c) = 0$. The *critical value* $P(c)$ plays a central role in polynomial dynamics, Smale’s mean value conjecture, and root-finding algorithms.

Classical Laguerre bounds involve the ratio $P'(z)/P(z)$. At a critical point, $P'(c) = 0$, so these bounds collapse to zero and give no information about $|P(c)|$. Our main result is a *non-trivial* lower bound on $|P(c)|$ that arises from the spectral structure of the partial-fraction decomposition of $1/P$.

2 The fiber–bridge identity

Theorem 2.1 (Fiber–bridge). *For any monic polynomial P of degree d and any $c \notin \{\alpha_j\}$:*

$$\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c) P'(\alpha_j)} = -\frac{1}{P(c)}. \quad (1)$$

Proof. The partial-fraction expansion of $1/P(z)$ is $\sum 1/((z - \alpha_j)P'(\alpha_j))$. Evaluating at $z = c$ (with $P(c) \neq 0$ generically) gives the result after the sign change $(c - \alpha_j) = -(\alpha_j - c)$. $\square$

Theorem 2.2 (Fiber vanishing). *If c is a critical point of P ($P'(c) = 0$), then:*

$$\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c)^2 P'(\alpha_j)} = 0. \quad (2)$$

Proof. Define $Q_2(c, z) = (P(z) - P(c))/(z - c)^2$, a polynomial of degree $d - 2$ in z (using $P'(c) = 0$ to ensure divisibility). Since $\deg Q_2 < d - 1$, the Lagrange interpolation identity gives $\sum Q_2(c, \alpha_j)/P'(\alpha_j) = 0$. Substituting $Q_2(c, \alpha_j) = -P(c)/(\alpha_j - c)^2$ and dividing by $-P(c)$ yields the result. $\square$

Corollary 2.3 (Root vanishing). $\sum 1/P'(\alpha_j) = 0$ (from the Lagrange identity applied to the constant 1).

3 Orthogonality structure

At a critical point c , define the spectral weight and distance vectors:

$$u_j = \frac{1}{(\alpha_j - c) P'(\alpha_j)}, \quad v_j = \alpha_j - c, \quad j = 0, \dots, d-1.$$

The three identities become:

Proposition 3.1 (Spectral orthogonality).

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum u_j v_j = \sum \frac{1}{P'(\alpha_j)} = 0, \quad (3)$$

$$\langle \mathbf{u}, \mathbf{1} \rangle = \sum u_j = -\frac{1}{P(c)}, \quad (4)$$

$$\langle \mathbf{u}, \mathbf{1}/\mathbf{v} \rangle = \sum \frac{u_j}{v_j} = \sum \frac{1}{(\alpha_j - c)^2 P'(\alpha_j)} = 0. \quad (5)$$

The vector $\mathbf{u}$ lies in the orthogonal complement of $\text{span}\{\mathbf{v}, (\mathbf{1}/\mathbf{v})_\perp\}$ in $\mathbb{C}^d$.

4 Main inequality

Theorem 4.1 (Laguerre–Pandrosion inequality). *Let P have degree d with roots $\alpha_0, \dots, \alpha_{d-1}$, and let c be a critical point. Define:*

- The spectral norm: $\Sigma^2(c) = \sum_{j=0}^{d-1} \frac{1}{|\alpha_j - c|^2 |P'(\alpha_j)|^2}$.
- The effective dimension: $\pi(c) = d - |\langle \hat{\mathbf{1}}, \hat{\mathbf{v}} \rangle|^2 - |\langle \hat{\mathbf{1}}, \widehat{(\mathbf{1}/\mathbf{v})_\perp} \rangle|^2$,
where $\hat{\mathbf{w}} = \mathbf{w}/\|\mathbf{w}\|$ and $(\mathbf{1}/\mathbf{v})_\perp$ is the component of $\mathbf{1}/\mathbf{v}$ orthogonal to $\mathbf{v}$.

Then:

$$\boxed{\frac{1}{|P(c)|^2} \leq \Sigma^2(c) \cdot \pi(c).} \quad (6)$$

Equivalently:

$$|P(c)| \geq \frac{1}{\sqrt{\pi(c) \cdot \Sigma^2(c)}}. \quad (7)$$

The effective dimension satisfies $\pi(c) \leq d - 2$ when $\mathbf{v}$ and $\mathbf{1}/\mathbf{v}$ are linearly independent.

Proof. By the Cauchy–Schwarz inequality with orthogonality constraints:

$$|\langle \mathbf{u}, \mathbf{1} \rangle|^2 \leq \|\mathbf{u}\|^2 \cdot \|\mathbf{1}_\perp\|^2,$$

where $\mathbf{1}_\perp$ is the projection of the all-ones vector $\mathbf{1}$ onto the orthogonal complement of $\text{span}\{\hat{\mathbf{v}}, \widehat{(\mathbf{1}/\mathbf{v})_\perp}\}$. The left side equals $1/|P(c)|^2$ by (4). The right side equals $\Sigma^2(c) \cdot \pi(c)$ by definition.

For the bound $\pi(c) \leq d - 2$: the all-ones vector $\mathbf{1}$ lives in $\mathbb{C}^d$. Projecting away the 2D subspace $\text{span}\{\hat{\mathbf{v}}, \widehat{(\mathbf{1}/\mathbf{v})_\perp}\}$ leaves a component of squared norm at most $d - 2$. $\square$

Remark 4.2 (Classical Laguerre fails at critical points). The classical Laguerre inequality bounds $|P'(z)|$ from above using root positions. At a critical point, $P'(c) = 0$, so the classical bound collapses. Theorem 4.1 provides a *complementary* bound: it gives a non-trivial lower bound on $|P(c)|$ precisely at critical points where the classical theory is silent.

5 Sharpness for $d = 3$

Theorem 5.1. *For $d = 3$, the Laguerre–Pandrosion inequality is sharp: equality holds for all critical points.*

Proof. For $d = 3$: $\mathbf{u}, \mathbf{v} \in \mathbb{C}^3$, and $\mathbf{u}$ is orthogonal to both $\mathbf{v}$ and $1/\mathbf{v}$. This leaves $\mathbf{u}$ in a 1-dimensional subspace (the orthogonal complement of $\{\mathbf{v}, (1/\mathbf{v})_\perp\}$ in $\mathbb{C}^3$). In a 1-dimensional space, Cauchy–Schwarz is an equality:

$$|\langle \mathbf{u}, \mathbf{1}_\perp \rangle|^2 = \|\mathbf{u}\|^2 \cdot \|\mathbf{1}_\perp\|^2. \quad \square$$

Remark 5.2. The sharpness for $d = 3$ is confirmed numerically: the ratio $|P(c)|^2/(1/(\Sigma^2\pi))$ equals 1.0000 across all tested polynomials.

6 Corollaries

Corollary 6.1 (Root-critical clustering). *If $|P(c)| < \varepsilon$ for some critical point c , then:*

$$\sum_{j=0}^{d-1} \frac{1}{|\alpha_j - c|^2 |P'(\alpha_j)|^2} \geq \frac{1}{(d-2)\varepsilon^2}.$$

In particular, at least one root must satisfy $|\alpha_j - c| \cdot |P'(\alpha_j)| \leq \sqrt{d-2} \cdot \varepsilon$.

Proof. From Theorem 4.1 with $\pi \leq d-2$: $1/\varepsilon^2 < 1/|P(c)|^2 \leq (d-2)\Sigma^2$, giving $\Sigma^2 \geq 1/((d-2)\varepsilon^2)$. The “in particular” follows from $\max |u_j|^2 \geq \Sigma^2/d$. $\square$

Corollary 6.2 (Smale ratio constraint). *For $P(z) = zR(z)$ with $R(0) = 1$ and critical c :*

$$S(c) = |R(c)| = \frac{|P(c)|}{|c|} \geq \frac{1}{|c| \sqrt{\pi(c) \cdot \Sigma^2(c)}}.$$

This gives a lower bound on the Smale ratio in terms of root geometry.

Corollary 6.3 (Near-root critical values). *If a critical point c is ε -close to a root ($|\alpha_j - c| = \varepsilon$ for some j), then:*

$$|P(c)| \geq \frac{\varepsilon |P'(\alpha_j)|}{\sqrt{(d-2)(\varepsilon^2 |P'(\alpha_j)|^2 \Sigma_{\neq j}^2 + 1)}}$$

where $\Sigma_{\neq j}^2$ is the spectral norm excluding the j -th root.

7 Numerical verification

d	Tests	Violations	min ratio	max ratio	Sharp?
3	10 000	0	1.0000	1.0000	Yes
4	15 000	0	1.0000	∞	No
5	20 000	0	1.0190	∞	No
6	25 000	0	1.0841	∞	No
7	30 000	0	1.1836	∞	No

Here “ratio” is $|P(c)|^2/(1/(\Sigma^2\pi))$, which is always ≥ 1 by Theorem 4.1. For $d = 3$, equality is achieved confirming sharpness.

8 Comparison with classical inequalities

Inequality	At regular z	At critical c
Classical Laguerre	$ P' ^2 \leq d \cdot (P'^2 - PP'')$	$0 \leq 0$ (trivial)
Bernstein–Markov	$ P' \leq d\ P\ _\infty$ (global)	$0 \leq d\ P\ _\infty$ (trivial)
Pandrosion energy	$P'^2 - PP'' = \sum Q_k^2 \geq 0$	$ P''P = \sum Q_k^2$ (known)
Laguerre–Pandrosion		$ \mathbf{P}(\mathbf{c}) \geq 1/\sqrt{\pi\Sigma^2}$ (new)

The Laguerre–Pandrosion inequality is the only known bound that gives *non-trivial information about $|P(c)|$ at critical points* using root geometry.

9 Proof of the fiber–bridge via partial fractions

For completeness, we give a self-contained proof of the fiber–bridge identity.

Proof of Theorem 2.1. The partial fraction expansion of $1/P(z)$ for distinct roots is:

$$\frac{1}{P(z)} = \sum_{j=0}^{d-1} \frac{1}{P'(\alpha_j)(z - \alpha_j)}.$$

Evaluating at $z = c$:

$$\frac{1}{P(c)} = \sum_{j=0}^{d-1} \frac{1}{P'(\alpha_j)(c - \alpha_j)} = - \sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c)P'(\alpha_j)}.$$

□

Proof of Theorem 2.2. Since $P'(c) = 0$, the polynomial $Q_2(c, z) = (P(z) - P(c))/(z - c)^2$ is well-defined and has degree $d - 2$. At each root α_j : $Q_2(c, \alpha_j) = (0 - P(c))/(\alpha_j - c)^2 = -P(c)/(\alpha_j - c)^2$. By the Lagrange identity for polynomials of degree $< d - 1$:

$$\sum_{j=0}^{d-1} \frac{Q_2(c, \alpha_j)}{P'(\alpha_j)} = 0.$$

Substituting and dividing by $-P(c) \neq 0$:

$$\sum_{j=0}^{d-1} \frac{1}{(\alpha_j - c)^2 P'(\alpha_j)} = 0.$$

□

References

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